# Preference Descriptions for Dynamic Personalization of Large Language Models

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Paper

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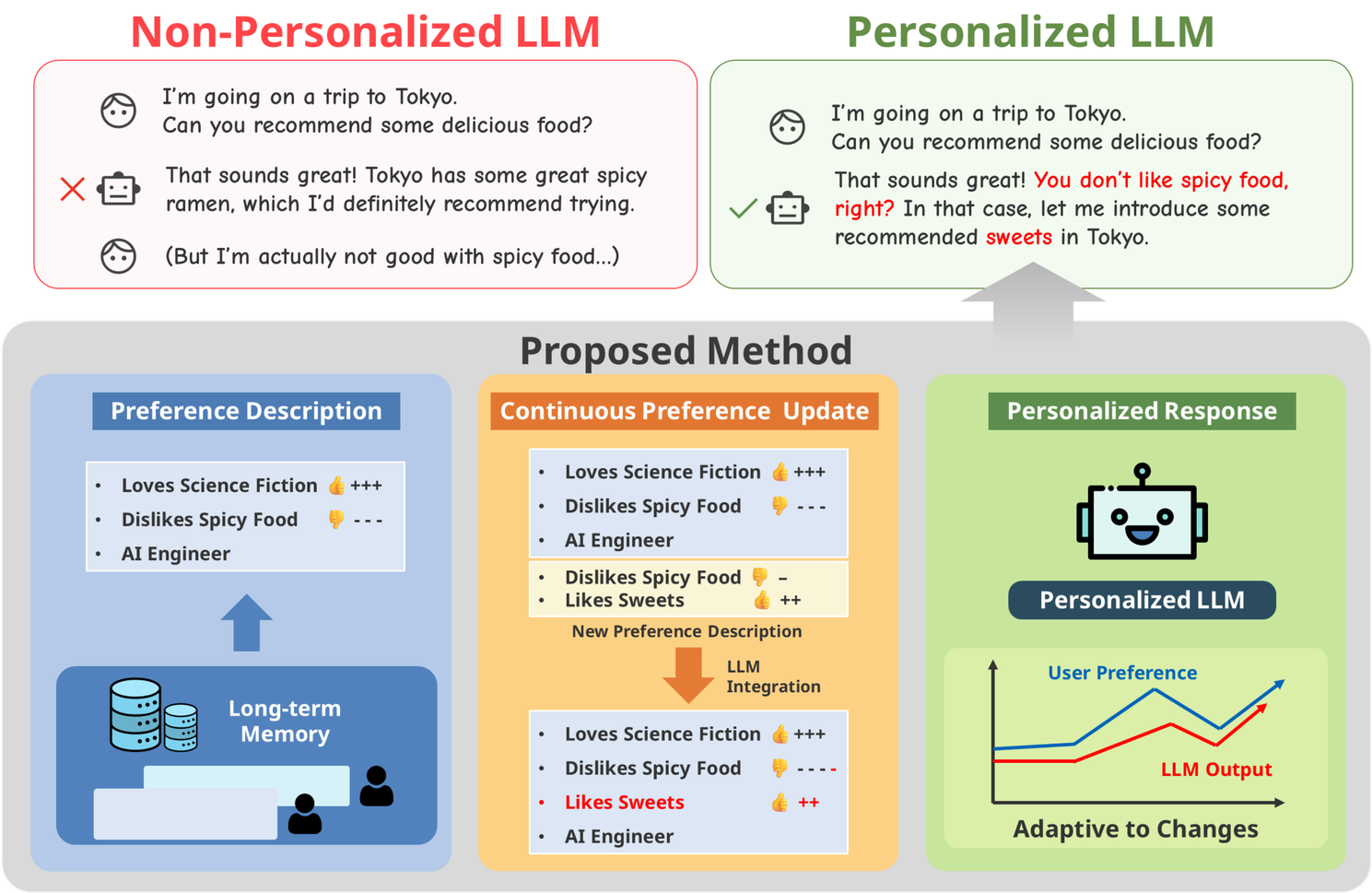
## Key Concepts

**Explicit preference descriptions with polarity & intensity, continuously updated at test time.**




### Preference Description

**Text :**
Natural language preference description

**Polarity :**
{-1,0,1}(Dislike / Neutral / Like)

**Intensity :**
{0–5}(Indifferent → Very Strong)

### Continuous Preference Update

- Aligns long-term preferences with recent interactions
- Updates preference description at test time using mini-batches

**Update Rule :** $L^* \leftarrow LLM(S_c, L^*, D_{\text{batch}})$

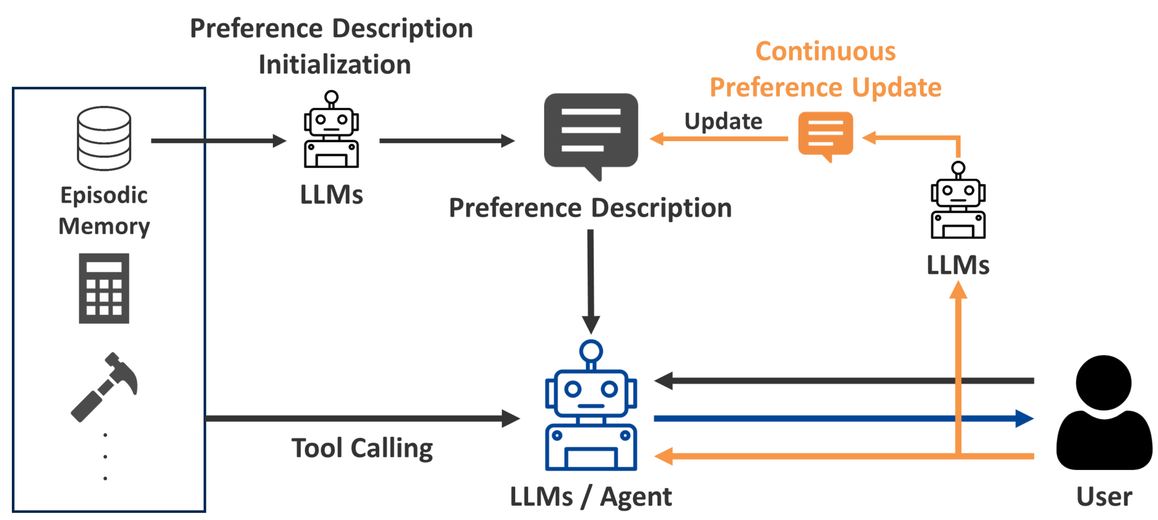
## Overall Architecture

### Continuous Personalization Framework

- Adapts to user preferences using long-term and short-term memory

### Components

- Episodic Memory : Stores past interactions
- Continuous Alignment : Updates preferences at test time




**Proposed Method Architecture**

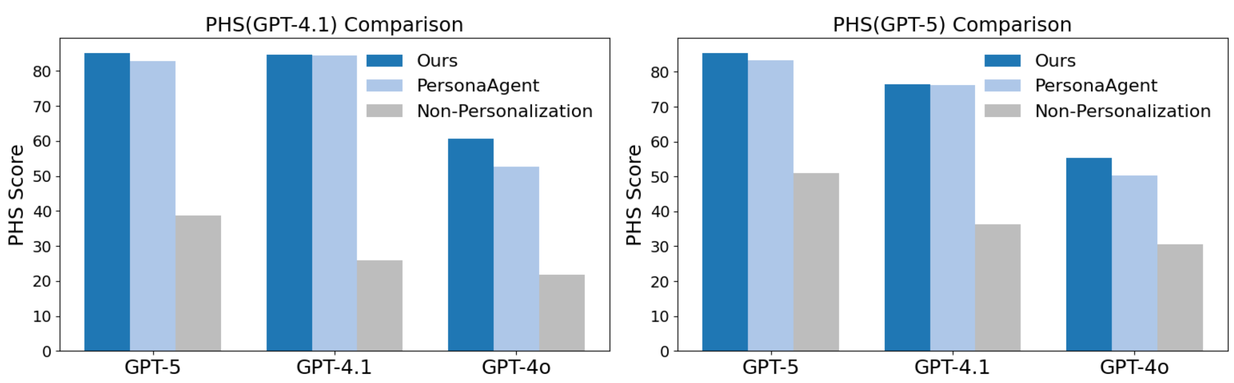
## Experiments & Results

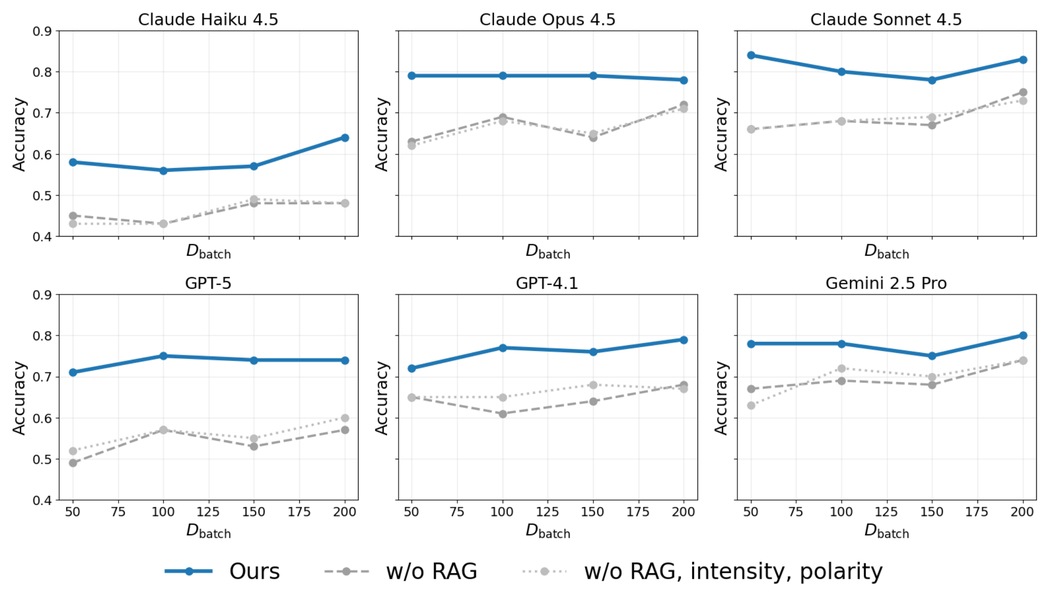
**Continuously updated preference representations improve preference alignment and long-term dialog accuracy.**

### Exp. 1 : Static Preference Injection

- Improves Persona Hit Score (PHS) across LLMs
- Responses reflect user preferences more explicitly and consistently
- Outperforms non-personalized and prompt-only baselines

Persona Hit Score (PHS) :
LLM-based evaluation of preference satisfaction (0–100)




### Exp. 2 : Continuous Preference Update

- Improves accuracy in long multi-turn dialogs
- Successfully adapts to preference drift over time
- Episodic memory (RAG) consistently boosts performance
- Moderate update intervals are more stable than frequent updates